Desk Watch

Screw & Fastener Guide



The Desk Watch case is printed in three parts - base, shell and tray (top plate) - and held together with small metal screws that bite directly into the PLA. This guide lists every screw you need, where it goes and how long it should be, so you can order the right pack before you start.

> **Which screws?**
> **M3** (8 in total): **4** to fix the base on the shell, and **4** to fix the Arduino UNO on the base.
> **M2 or M1.7**: everything else (the tray / top plate, the LCD and all Grove modules).

What to buy

Screw	Qty	Used for
M2 x 5 mm, pan head	14	LCD (5) and the four Grove modules (9)
M2 x 6 mm, pan head	4	Top plate on the case (4 corners)
M3 x 6 mm, pan head	4	Arduino UNO on the base (4 posts)
M3 x 12 mm, countersunk (flat head)	4	Base on the case (4 corners)
Cable ties 2.5 x 100 mm (optional)	4	Cable routing in the base

Order about 20 % more than listed: M2 screws are tiny and easy to lose. If you only have M2 x 5, it works everywhere an M2 is listed. These counts are for the full build with the dial and the button; without them, you need 4 fewer M2 x 5.

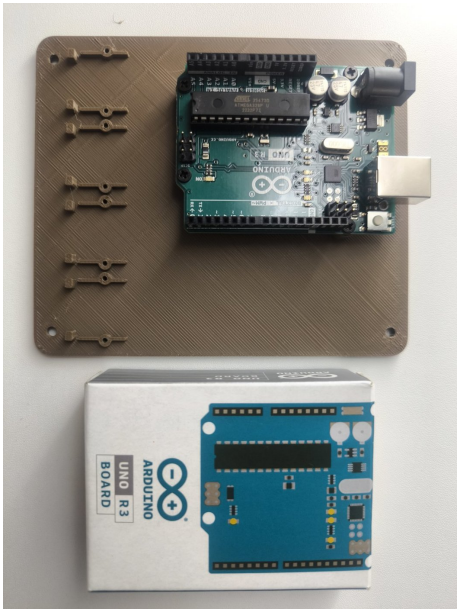
Tip from the builder: M1.7 x 6 mm sharp-point screws

Sharp-point (self-tapping) M1.7 x 6 mm screws also work very well in place of the M2 screws, especially since PLA is not threaded. Their heads still fit the recesses of the top plate and the holes of the Grove modules (about 2 mm) leave them a little clearance. For the four plate corners, keep M2 x 6: they have a larger core and hold better when you pull on the plate.

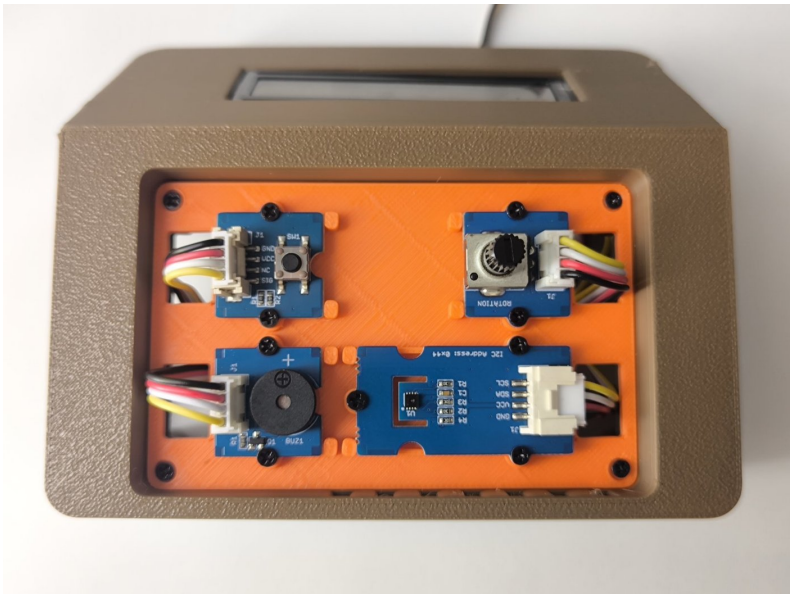
The pilot holes in the model are sized for M2 and M3 self-tapping screws. With M1.7, test on one module position first: the thread may bite slightly less in a hole made for M2.

Where each screw goes

#	Fastening	Qty	Screw	Notes
1	Arduino UNO on the base, 4 posts	4	M3 x 6 (M3 x 8 possible, 10 mm max.)	Posts 6 mm wide, 2.6 mm pilot hole, 6 mm deep, at the four standard UNO mounting holes. A screw longer than 10 mm comes out under the base.
2	Base on the case, 4 corners	4	M3 x 12 countersunk, screwed from below (M3 x 10 possible)	90 degree countersink, 6.5 mm wide. Pilot hole 11.8 mm deep: 14 mm max.
3	LCD on the posts of the case	5	M2 x 5, pan or round head	5 of the 6 holes are used: the post of the left hole was removed on purpose.
4	Top plate on the case rim, 4 corners	4	M2 x 6, round head max. 4 mm (M2 x 5 possible)	Recess 4.4 mm wide, 1.2 mm deep: do not use countersunk screws. 6.8 mm max. under the head.
5	Grove modules on the top plate	9	M2 x 5 (6 mm max.)	SHT31: 3 screws, buzzer: 2, dial: 2, button: 2. Posts 4.6 mm, pilot hole 1.7 mm.
6	Cable ties (optional)	4	2.5 x 100 mm	4 eyelets in the base.



1 - The UNO sits on M3 posts of the base (photo of an earlier base: the current base has four posts).



5 - Each Grove module is screwed on the top plate through its two holes (3 for the SHT31).

Suggested assembly order

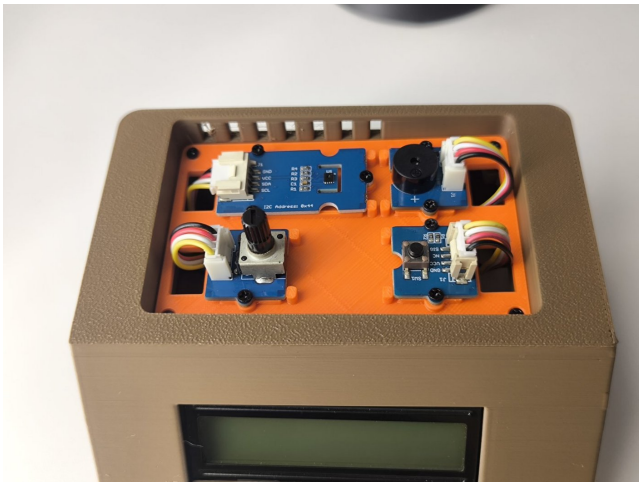
1. **Base.** Fix the UNO on its four posts with four M3 x 6 screws. Do not overtighten: PLA only needs a light grip.
2. **LCD.** Screw the display on the five posts of the case with M2 x 5.
3. **Top plate.** Fix each Grove module with its M2 x 5 screws, plug the Grove cables, then screw the plate on the case rim with four M2 x 6.
4. **Close.** Lay the cables in the channels of the base, then fix the base from below with four M3 x 12 countersunk screws.

Good to know

Self-tapping in PLA. All pilot holes are made for screws that cut their own thread in the plastic. Drive the screw in slowly, and stop as soon as it is snug.

M2.5 screws. If you only have M2.5 x 8 or x 12, they fit the UNO (x 8) and the base-to-case corners (x 12), but the pilot holes there are 2.6 mm - sized for M3. For M2.5 you would need about 2.1 mm, so a loose grip is likely.

Do your first print test on the base. Check that your UNO and your M3 screws fit, then print the top plate, then the case.



Top plate with the four Grove modules in place.



All parts before assembly.

What is verified and what is not

Lengths and engagement depths come from measurements on the 3D model (case, base and top plate). Screw head diameters (about 4 mm for M2, 6 mm for M3) are typical values: check them on your own screws. The four UNO posts and their 2.6 mm pilot holes were measured on the current base file; the holes of your UNO board are assumed to be the standard 3.2 mm. The third hole of the SHT31 was read from a photo (about 1 mm of accuracy). The M1.7 x 6 tip comes from the author's own build.

Model files: MakerWorld, Desk Watch by Pr. BELAMINE (makerworld.com/models/3451026).